

Product Requirements Document: AI-Powered Landing Page Builder MVP

1. Project Overview:

The goal of this project is to develop an AI-powered application that empowers users to effortlessly create highly customized landing pages for any niche. Through an intuitive, conversational chat interface (akin to ChatGPT), users will describe their product, service, or personal brand, and the AI will intelligently generate a tailored landing page. The application will provide extensive customization options for various sections, allow for rich media integration, and offer flexible publishing methods, including direct download and integration instructions. The core focus is on democratizing landing page creation, making it accessible and efficient for individuals and small businesses.

2. Level: Easy to Medium

3. Type of Project: AI Development, Web Development, UI/UX, Personalization

4. Skills Required:

- **Backend & AI:** Python (for AI logic and API integration), **FastAPI**, familiarity with large language models (LLMs) and their APIs (e.g., OpenAI API, Gemini API).
- **Frontend:** JavaScript/TypeScript, **Next.js**, **Tailwind CSS**, HTML5, CSS3.
- **Data Handling:** Basic understanding of data storage (e.g., **Supabase for its flexible schema capabilities and real-time features**, or other NoSQL options for user preferences and generated content drafts if persistence is added later).
- **Deployment:** Understanding of web hosting environments (e.g., Netlify, Vercel, Firebase Hosting) and static site generation concepts for the downloadable output.
- **Image/Video Processing:** Basic understanding of how to handle image/video uploads and embedding.

5. Key Features (Minimum Viable Product - MVP):

Milestone 1: Conversational AI & Core Content Generation

- **5.1. AI-Powered Landing Page Generation Engine:**
 - **5.1.1. Chat Interface:** A clean, intuitive chat UI (similar to ChatGPT) where users input prompts describing their landing page needs (e.g., "Create a landing page for my new eco-friendly water bottle," "Generate a personal landing page for a freelance graphic designer").
 - **5.1.2. Initial Draft Generation:** The AI will interpret the user's prompt and generate a complete initial draft of a landing page. This draft will include placeholder content for each standard section (Hero, Features, Testimonials, CTA, Footer). The AI will attempt to generate relevant headlines, sub-headlines, feature points, dummy testimonials, and CTA text based on the niche described.
 - **5.1.3. Niche Customization:** The AI will be trained or prompted to generate content highly relevant to the specified niche, ensuring the initial output is a strong starting point.

Milestone 2: Section-Specific Customization & Media Integration

- **5.2. Section Customization Controls:** For each generated section, users will have dedicated editing controls.
 - **5.2.1. Hero Section Customization:**
 - **Customization Options:**
 - **Headline Text:** Editable text input, font size, bold/italic options.
 - **Sub-headline Text:** Editable text input, font size.
 - **Call-to-Action (CTA) Button:** Editable text, link URL input.
 - **Background:** Option to upload an image, select a solid color, or embed a video (via URL from YouTube/Vimeo).
 - **Overlay:** Color picker for an overlay tint on background images/videos, opacity slider.
 - **Elements:** Text, Image, Video, Button.
 - **5.2.2. Features Section Customization:**
 - **Customization Options:**
 - **Section Title:** Editable text.
 - **Individual Feature Blocks:** Add/remove feature blocks.
 - **Feature Title:** Editable text for each feature.
 - **Feature Description:** Editable multi-line text for each feature.
 - **Icon/Image per Feature:** Option to select an icon from a pre-defined library (e.g., Lucide React, Font Awesome basic set) or upload a small image.
 - **Elements:** Text, Images/Icons.

- **5.2.3. Testimonials Section Customization:**
 - **Customization Options:**
 - **Section Title:** Editable text.
 - **Individual Testimonial Blocks:** Add/remove testimonial blocks.
 - **Testimonial Quote:** Editable multi-line text.
 - **Author Name:** Editable text.
 - **Author Title/Company:** Editable text.
 - **Author Image:** Option to upload an image for the author's avatar.
 - **Elements:** Text, Images.
- **5.2.4. Call to Action (CTA) Section Customization (Main Page CTA):**
 - **Customization Options:**
 - **CTA Headline:** Editable text.
 - **CTA Sub-text:** Editable text.
 - **CTA Button Text:** Editable text.
 - **CTA Button Link:** URL input.
 - **Elements:** Text, Button.
- **5.2.5. Footer Section Customization:**
 - **Customization Options:**
 - **Copyright Text:** Editable text.
 - **Navigation Links:** Add/edit/remove simple text links with URLs (e.g., "Privacy Policy," "Terms of Service").
 - **Social Media Icons:** Pre-defined icons (e.g., Facebook, Twitter, LinkedIn) with URL inputs.
 - **Elements:** Text, Links, Icons.
- **5.2.6. Social Proof Section Customization:**
 - **Customization Options:**
 - **Section Title:** Editable text.
 - **Multiple Testimonials:** Ability to add various types of testimonials (text, video embeds, links to case studies). Each testimonial block will have fields for content, author, title/company, and media (image/video).
 - **Client Logos:** Add/remove client logos with optional short blurbs or links.
 - **Rating Aggregators:** Input fields for average rating and number of reviews, possibly a link to the review site.
 - **Awards and Press Mentions:** Add/remove entries for awards or press mentions, including title, organization, and link.
 - **Elements:** Text, Images (for logos, testimonial videos, awards), Video (embeds), Links.
- **5.2.7. Pricing Section Customization (if applicable):**
 - **Customization Options:**
 - **Section Title:** Editable text.
 - **Pricing Tiers:** Add/remove pricing cards. For each card: plan name,

- price, billing frequency, list of features included, CTA button text, CTA button link.
 - **Feature Comparisons:** Option to display a comparison table with features across different tiers.
 - **Pricing-related FAQs:** A dedicated sub-section for common questions about pricing plans.
 - **Elements:** Text, Numbers, Buttons, Tables.
- **5.2.8. Use Cases/Target Audience Section Customization:**
 - **Customization Options:**
 - **Section Title:** Editable text.
 - **Individual Use Cases:** Add/remove blocks detailing specific use cases or target audience segments. Each block will have a title, description, and optional image/icon.
 - **Elements:** Text, Images/Icons.
- **5.2.9. FAQ Section Customization:**
 - **Customization Options:**
 - **Section Title:** Editable text.
 - **Comprehensive List of Q&A:** Add/remove individual question-and-answer pairs. Each pair will have fields for the question and a multi-line answer.
 - (Future: Accordion/toggle functionality for answers).
 - **Elements:** Text.
- **5.3. General Media Elements:**
 - **5.3.1. Image Upload:** Users can upload JPG, PNG, GIF images. The system will handle basic resizing/optimization for web. **Support drag-and-drop functionality for image uploads.**
 - **5.3.2. Video Embedding:** Support for YouTube and Vimeo video URLs, which will be embedded via iframes. **Support drag-and-drop functionality for video file uploads (if direct file upload is supported, otherwise for URL input).**
 - **5.3.3. Text Editing:** Basic rich text editing (bold, italic, links) for all editable text fields.

Milestone 3: AI-Powered Content Enhancements (Gemini API Integration)

Leveraging the Gemini API, the application provides intelligent assistance for copywriting and optimization.

- **AI Copy Suggestions ✨:** Leverage Gemini API to get intelligent suggestions for headlines, sub-headlines, feature descriptions, and CTA button text based on your content and niche.
- **AI SEO Optimization ✨:** Utilize Gemini API to generate optimized meta titles and descriptions for improved search engine visibility.

Milestone 4: Design & Publishing

- **5.4. Global Design Customization:**
 - **5.4.1. Color Palette:**
 - **Primary Brand Color:** Main color for accents, buttons, key headings.
 - **Secondary Accent Color:** Complementary color for secondary elements.
 - **Text Colors:** Separate controls for heading text color and body text color.
 - **Background Colors:** Controls for overall page background, and potentially distinct background colors for alternating sections (e.g., light/dark bands).
 - **Button Colors:** Specific controls for button background and text color.
 - **Mechanism:** User-friendly color pickers (HEX, RGB input).
 - **5.4.2. Typography:**
 - **Font Family Selection:** A curated list of popular, performant web fonts (e.g., from Google Fonts). Users select a primary font for headings and a secondary font for body text.
 - **Font Sizing:** Sliders or dropdowns for adjusting base font sizes for headings (H1-H4) and body text.
 - **5.4.3. Themes:**
 - **Pre-designed Themes:** Offer 3-5 distinct pre-designed themes (e.g., "Clean & Modern," "Bold & Energetic," "Classic Professional"). Applying a theme will set initial colors, fonts, and potentially section layouts, which users can then further customize.
- **5.5. Publishing Options:**
 - **5.5.1. Live Preview:** An instant, interactive preview pane that updates in real-time as the user makes changes. This preview should be fully responsive.
 - **5.5.2. Download as ZIP File:**
 - Generates a ZIP archive containing all necessary files: index.html, style.css, script.js (for any interactive elements), and an images folder containing all uploaded assets.
 - The HTML will be clean, semantic, and well-structured.
 - CSS will be well-organized and modular, ideally using Tailwind CSS classes for consistent styling.
 - All assets will be correctly linked within the generated files.
 - **5.5.3. Integration Instructions for User's Website:**
 - Provide clear, step-by-step instructions on how to upload the ZIP file to common hosting providers (e.g., cPanel) or integrate the HTML/CSS into popular website builders/CMS platforms (e.g., WordPress custom pages, Squarespace code blocks).
 - Include a basic HTML snippet that users can copy-paste into an existing page if they only want a specific section.
 - **5.5.4. Publish Online (Vercel Integration - MVP):**

- Provide a one-click deployment option to Vercel. This would require the user to connect their Vercel account (or similar simple authentication flow) and grant necessary permissions.
- The system would then deploy the generated landing page as a static site on Vercel, providing the user with a live URL.
- *(Future thought: Option to connect custom domains.)*

6. User Flow (High-Level):

1. **Onboarding:** User lands on the app, sees a brief explanation of how it works.
2. **Start New Page:** User clicks "Create New Landing Page."
3. **AI Interaction:** User types a prompt describing their landing page (niche, purpose, key message).
4. **Initial Generation:** AI processes the prompt and presents an initial draft of the landing page in the preview pane, with content populated in sections.
5. **Customization:**
 - User navigates to different sections (Hero, Features, Social Proof, etc.) via a sidebar or tabbed interface.
 - User edits text, uploads images/videos (including drag-and-drop), adjusts section-specific settings.
 - User accesses global design settings (colors, fonts, themes) to refine the overall look.
6. **Preview:** User continuously reviews changes in the responsive live preview.
7. **Publish/Download:** User clicks "Publish" or "Download," selects the desired option (ZIP download, publish online to Vercel, view integration instructions).

7. Technical Considerations (High-Level):

- **Frontend: Next.js** for the frontend framework, leveraging its capabilities for static site generation (SSG) or server-side rendering (SSR) if dynamic content is introduced later. **Tailwind CSS** will be used for efficient and responsive styling.
- **Backend (for AI & Deployments): FastAPI** for the lightweight Python backend, handling requests to the LLM API, managing file uploads, and interacting with deployment services like Vercel's API.
- **AI Integration:** Securely call the chosen LLM API (e.g., Gemini API) for content generation based on user prompts. For the MVP, data streamlining to the AI/NLP should be minimal, primarily involving direct passing of the user's conversational prompt and context, without extensive pre-processing or complex feature engineering. Ensure proper API key management (handled by the Canvas environment).
- **Asset Management:** Simple file upload handling for images (could use a cloud storage solution like Firebase Storage or **Supabase Storage** in a full app, but for MVP, local

temporary storage on the backend before zipping or direct client-side upload to a temporary bucket could work). Implementing drag-and-drop will involve frontend JavaScript event listeners.

- **HTML/CSS/JS Generation:** The frontend application will dynamically construct the HTML, CSS (using Tailwind classes for efficiency and responsiveness), and minimal JavaScript based on user customizations, which can then be bundled into a ZIP file.
- **Database (Optional for MVP, but for persistence):** For saving user projects and settings, **Supabase is a strong option, leveraging its PostgreSQL backend for structured data and its Storage service for files. It can be used in a schema-flexible (NoSQL-like) manner with JSONB columns for specific data structures.**
- **Vercel Integration:** Requires using Vercel's API for programmatic deployment. This would involve handling API keys/tokens securely on the backend.
- **Security:** Implement industry best practices for web application security, including but not limited to: input validation, secure API key handling, CORS policies, secure authentication (if user accounts are introduced), and protection against common web vulnerabilities (e.g., XSS, CSRF).

8. Success Metrics:

- Number of landing pages generated.
- User satisfaction with the customization options.
- Successful downloads/publishing attempts (including Vercel deployments).
- Time taken to create a landing page from scratch.

This PRD provides a solid foundation for your MVP. Remember to iterate based on user feedback and focus on getting the core AI generation and essential customization features right first!